

Titanic Dataset - Exploratory Data Analysis (EDA) Detailed Report

This report explains the Python EDA project created on the Titanic dataset. The project uses Python libraries like Pandas, NumPy, Matplotlib, and Seaborn to analyze passenger survival data and generate different visualizations.

1. Importing Libraries

The code first imports important Python libraries required for data analysis and visualization.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

2. Loading Dataset

The Titanic CSV file is loaded into a DataFrame using Pandas. A DataFrame stores tabular data in rows and columns.

```
df = pd.read_csv('titanic.csv')
```

3. Customizing Chart Appearance

The project defines custom color palettes and chart settings to improve visualization quality and readability.

```
NAVY = '#1B2A4A'
TEAL = '#2E86AB'
CORAL = '#E84855'
```

4. Survival Rate by Gender

This section calculates the percentage of male and female passengers who survived and displays the result using a bar chart.

```
gender_survival = df.groupby('Sex')['Survived'].mean() * 100
ax.bar(gender_survival.index, gender_survival.values)
```

5. Survival Rate by Passenger Class

The code compares survival percentages among 1st, 2nd, and 3rd class passengers.

```
class_surv = df.groupby('Pclass')['Survived'].mean() * 100
```

6. Age Distribution Analysis

Histograms are used to compare age distributions between survivors and non-survivors.

```
ax.hist(age_died, bins=25)
ax.hist(age_survived, bins=25)
```

7. Heatmap Analysis

A heatmap visualizes survival percentage based on passenger class and gender.

```
pivot = df.pivot_table(values='Survived',  
index='Pclass',  
columns='Sex',  
aggfunc='mean')
```

8. Fare Distribution

Boxplots are used to analyze ticket fare variations across different passenger classes.

```
ax.boxplot(data_by_class, patch_artist=True)
```

9. Embarkation Port Analysis

The analysis compares survival rates based on passenger boarding ports.

```
port_data = df.groupby('Embarked')['Survived'].agg(['mean', 'count'])
```

10. Statistical Summary

The code prints useful statistics like total passengers, survival percentage, missing values, and fare statistics.

```
print(df.groupby('Sex')['Survived'].agg(['sum', 'mean', 'count']))
```

Conclusion

The Titanic EDA project demonstrates how Python can be used for real-world data analysis. Using visualizations and statistical analysis, the project identifies patterns in passenger survival based on gender, age, passenger class, fare, and embarkation port. The project is useful for learning data analysis, data visualization, and machine learning preprocessing concepts.